



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Mini Project

HOMESTAY XYZ BOOKING SYSTEM

Subject: SECJ2013- Data Structure and Algorithm

Session-Sem: 2024/2025-1

Section: 02

Group: G7

Video Presentation: <https://youtu.be/C11OJ2OIeTw?si=Uf4xgzpBQHXLr4oI>

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1 The synopsis and objective

1.1 Synopsis

Homestay XYZ has been facing difficulties in managing its process due to the inefficiency of the current booking system. This brings a lot of problems to customers and staff when using a poor booking system. For example, customers overbooking the room lack the ability to check room availability, make payments, and cancel bookings. This is very inconvenient for customers and also will decrease the customer's satisfaction when using a poor booking system. On the other hand, the inefficiency of the Homestay XYZ booking system also makes staff members face challenges in managing or viewing booking requests of the customers. Besides that, the staff members also find it difficult to track the availability of rooms in Homestay XYZ. They cannot track the availability in real-time information on the availability or capacity of homestays. This also brings inconvenience to staff members when using the system.

Therefore, a new Homestay XYZ booking system is being developed in this mini project. In this booking system, there are two main users, which are customers and staff. The booking system for Homestay XYZ is suitable for the application of the queue concept. This is because the booking system will manage and process customers requests in the order they are received. Therefore, the Homestay XYZ booking system will handle booking more fairly as each request is processed one at a time in a sequential manner. Moreover, the queue concept is chosen because the queue follows the FIFO (First In, First Out) principle. By processing the oldest booking request first, the booking system will guarantee that no customer's booking requests are skipped.

With the development of the new booking system in Homestay XYZ, customers are able to make bookings, make payments, cancel bookings, and check room availability more efficiently and smoothly. For staff members, the system allows them to view and manage booking requests more efficiently and track room availability in real time.

1.2 Objective

- To develop a booking system that uses a queue data structure and applies the FIFO principle.
- To manage customer booking requests efficiently.
- To allow customers to make booking requests.
- To allow customers to cancel their booking.
- To allow customers to check room availability or capacity of homestay.
- To allow staff to view all the booking requests of customers.
- To allow staff to check the availability or capacity of homestay.
- To allow staff to process bookings in FIFO order.

2 System requirement, analysis & design (ADT by class diagram)

2.1 System Use Case Diagram

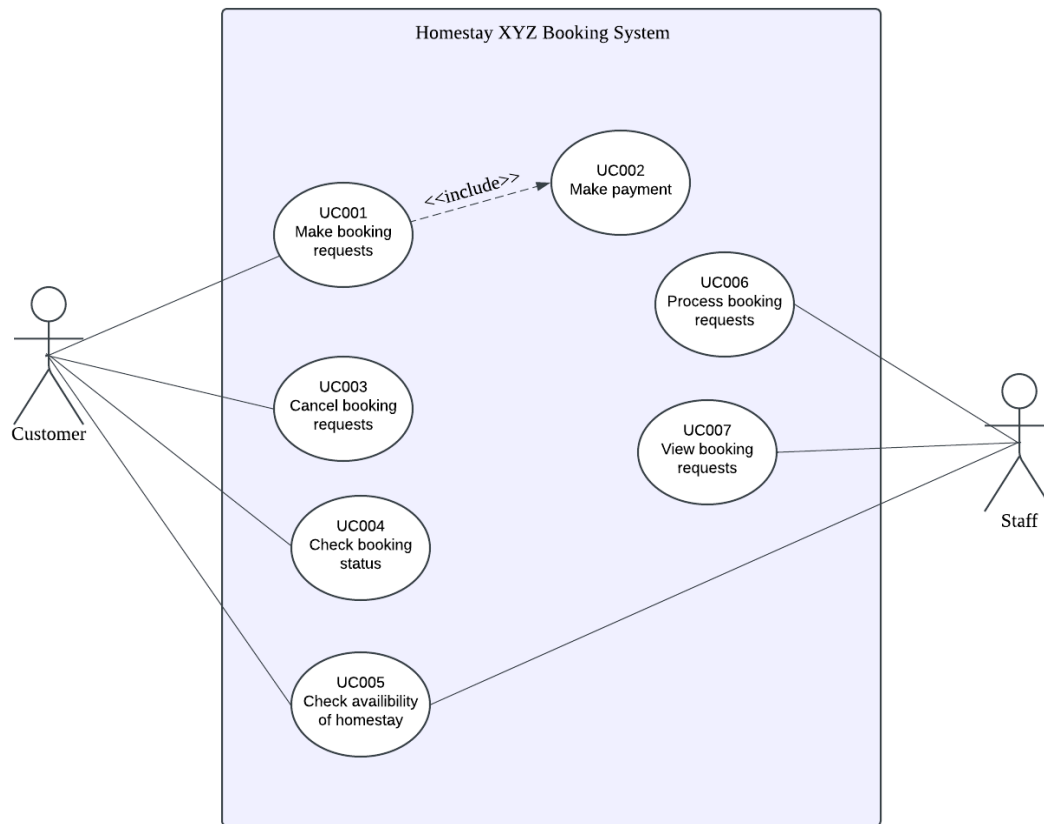


Figure 1.1 Use case diagram for homestay XYZ booking system

2.2 System Class Diagram

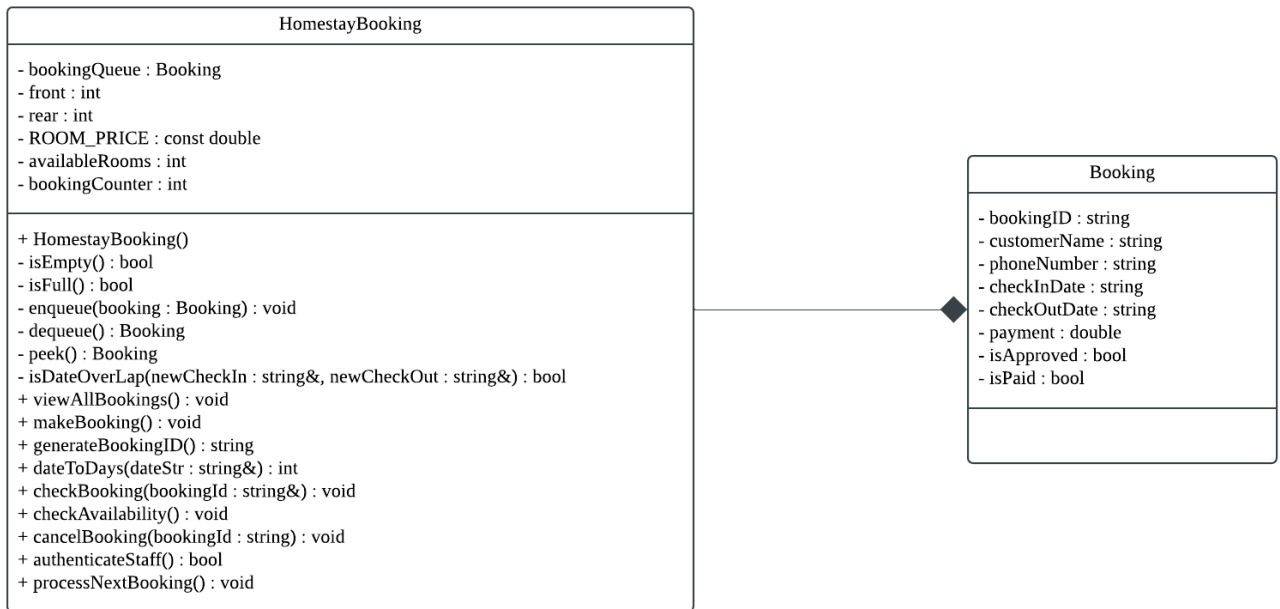


Figure 1.2 Class diagram for homestay XYZ booking system

3 Development activities and coding in C++

```
Welcome to Homestay XYZ
1. Customer
2. Staff
3. Exit
```

Figure 3.1 : Main Menu of Homestay XYZ

Figure 3.1 is the main menu of the Homestay XYZ booking system. Users allow keying in an integer value from 1 to 3. From the main menu, the customer will choose option 1 to access the customer menu and staff will choose option 2 to access the staff menu. If users did not want to use the booking system, they can choose option 3 to exit the Homestay XYZ booking system.

```
Customer Menu
1. Make new booking
2. Check booking status
3. Cancel booking
4. Check room availability
5. Return to main menu
Enter choice: |
```

Figure 3.2 : Customer Menu

The customer allows keying in an integer value from 1 to 5 in the customer menu. If the customer chooses option 1, the system will let the customer make a new booking. For option 2, the customer can check their booking status. For option 3, the customer can cancel the booking. For option 4, the customer can check the room availability of Homestay XYZ. For option 5, the customer will be returned to the main menu of the Homestay XYZ booking system.

```
Enter choice: 1

Enter customer name: Tan
Enter phone number: 01125668992
Enter check-in date (DD/MM/YYYY): 12/01/2025
Enter check-out date (DD/MM/YYYY): 14/01/2025

Booking Details:
Check-in: 12/01/2025
Check-out: 14/01/2025
Number of days: 2
Price per night: RM100
Total payment: RM200

Proceed with payment? (Y/N): y
```

Figure 3.3 : Customer Make a New Booking

```
Enter choice: 1

Enter customer name: Joelle
Enter phone number: 0123456789
Enter check-in date (DD/MM/YYYY): 12/01/2025
Enter check-out date (DD/MM/YYYY): 14/01/2025
Sorry, the homestay is already booked for these dates.
```

Figure 3.4 : Other Customer Make Booking with same dates

When the customer keyed in the number 1, the booking system will let the customer fill in customer information such as name and phone number. After that, the customer needs to fill in the check-in date and check-out date for booking the homestay. Then, the system will automatically show the booking details such as check-in date, check-out date, total number of days booked in a homestay, price per night and the total payment. The customer will proceed with payment when the customer keyed in "y" / "Y." The homestay will be full in a day when there are a total of 3 bookings made by different customers. So, when the new customer chooses to book for the same date after 3 bookings, the system will prompt, "Sorry, the homestay is already booked for these dates."

```
Proceed with payment? (Y/N): y
```

```
Payment successful!  
Booking confirmed!  
Your booking ID: A1  
-----
```

```
Receipt:  
Booking ID: A1  
Customer Name: Tan  
Check-in: 12/01/2025  
Check-out: 14/01/2025  
Amount Paid: RM200  
Status: Confirmed  
-----
```

Figure 3.5 : Customer Make Payment and the Detail of Booking

The booking of a homestay will be confirmed after the customer makes payment successfully and the booking ID and receipt will be generated.

```
Customer Menu  
1. Make new booking  
2. Check booking status  
3. Cancel booking  
4. Check room availability  
5. Return to main menu  
Enter choice: 2  
Enter booking ID: A1  
  
Booking found!  
Customer Name: Tan  
Check-in: 12/01/2025  
Check-out: 14/01/2025  
Status: Approved  
Payment: RM200
```

Figure 3.6 : Customer Check Booking Status

The customer can check their booking status by entering the number 2 and their booking ID. When the booking ID is found, the system will display the booking details of the customer.


```
Customer Menu
1. Make new booking
2. Check booking status
3. Cancel booking
4. Check room availability
5. Return to main menu
Enter choice: 3
Enter booking ID: A1
Booking cancelled successfully!
```

Figure 3.7 : Customer Cancel Booking

The customer can check their booking by entering the number 3 and their booking ID. When the booking ID is found, the system will cancel the booking of the customer.

```
Customer Menu
1. Make new booking
2. Check booking status
3. Cancel booking
4. Check room availability
5. Return to main menu
Enter choice: 4

Enter check-in date (DD/MM/YYYY): 12/01/2025
Enter check-out date (DD/MM/YYYY): 15/01/2025

Availability Status for 12/01/2025 to 15/01/2025:
-----
Status: NOT AVAILABLE
The homestay is already booked for these dates.
-----
```

Figure 3.8 : Customer Check Room Availability of Homestay XYZ

The customer can check the room availability by entering the number 4. The system will ask the customer to enter the check-in date and check-out date. When the dates overlap with customers that have already booked the homestay, the system will prompt, “The homestay is already booked for these dates.”

```
Customer Menu
1. Make new booking
2. Check booking status
3. Cancel booking
4. Check room availability
5. Return to main menu
Enter choice: 5

Welcome to Homestay XYZ
1. Customer
2. Staff
3. Exit
```

Figure 3.9 : Customer Return to Main Menu

The customer can return to the main menu of Homestay XYZ by entering the number 5. After that, the customer successfully returned to the main menu.

```
=== Homestay Booking System ===
1. Customer
2. Staff
3. Exit
Enter your choice: 2
Enter staff password: 123

=== Staff Menu ===
1. View All Bookings
2. Check Room Availability
3. Process Next Booking
4. Return to Main Menu
```

Figure 3.10 : Staff Access to the Staff Menu

The staff have to key in the number 2 and the staff password when they want to access the staff menu of Homestay XYZ. There are four options in the staff menu which are view all bookings, check room availability, process next booking and return to the main menu of Homestay XYZ.

```
=== Staff Menu ===
1. View All Bookings
2. Check Room Availability
3. Process Next Booking
4. Return to Main Menu
Enter your choice: 1

All Confirmed Bookings:
-----
Booking ID: A1
Customer Name: Tan
Phone: 01128774335
Check-in: 11/11/2025
Check-out: 12/11/2025
Payment: RM100
```

Figure 3.11 : Staff View All Bookings of the Customer

The staff can view all the bookings of the customers by entering the number 1 in the staff menu. The system will display all the confirmed booking details of the Homestay XYZ.

```
=== Staff Menu ===
1. View All Bookings
2. Check Room Availability
3. Return to Main Menu
Enter your choice: 2

Enter check-in date (DD/MM/YYYY): 11/11/2025
Enter check-out date (DD/MM/YYYY): 12/11/2025

Availability Status for 11/11/2025 to 12/11/2025:
-----
Status: NOT AVAILABLE
The homestay is already booked for these dates.
-----
```

Figure 3.12 : Staff Check Room Availability of Homestay XYZ

The staff can check room availability at Homestay XYZ by entering the number 3 in the staff menu. The availability status of the homestay will show “NOT AVAILABLE” when the homestay is already booked or bookings are full for the selected date.

```

=== Staff Menu ===
1. View All Bookings
2. Check Room Availability
3. Return to Main Menu
Enter your choice: 2

Enter check-in date (DD/MM/YYYY): 13/12/2025
Enter check-out date (DD/MM/YYYY): 14/12/2025

Availability Status for 13/12/2025 to 14/12/2025:
-----
Status: AVAILABLE
Number of nights: 1
Price per night: RM100
Total price: RM100

Would you like to make a booking? (Y/N): y

Enter customer name: Lew
Enter phone number: 0187689334
Enter check-in date (DD/MM/YYYY): 13/12/2025
Enter check-out date (DD/MM/YYYY): 14/12/2025

Booking Details:
Check-in: 13/12/2025
Check-out: 14/12/2025
Number of days: 1
Price per night: RM100
Total payment: RM100

Proceed with payment? (Y/N): y

Payment successful!
Booking confirmed!
Your booking ID: A2
-----
Receipt:
Booking ID: A2
Customer Name: Lew
Check-in: 13/12/2025
Check-out: 14/12/2025
Amount Paid: RM100
Status: Confirmed
-----

```

Figure 3.13 : Staff Check Room Availability of Homestay XYZ

When the availability of Homestay XYZ is available for the selected date, the availability status of the homestay will show “AVAILABLE” and display the details of the booking information. The staff will help the customer to book the homestay if the staff keyed in the “Y” or “y” when the system asks the staff to make a new booking. Therefore, the staff has to enter the information of the customers, check-in and check-out dates for the homestay. The booking of a homestay will be confirmed after the customer makes payment successfully and the booking ID and receipt will be generated.

```
=== Staff Menu ===
1. View All Bookings
2. Check Room Availability
3. Process Next Booking
4. Return to Main Menu
Enter your choice: 3

Processing Booking ID: A1
Customer Name: Tan
Check-in: 11/11/2025
Check-out: 12/11/2025
Status: Approved
Do you want to cancel this booking? (Y/N): y
Booking ID: A1 has been cancelled.
```

Figure 3.14 : Staff Process Next Booking

The staff process for the next booking involves handling the first booking in the queue (FIFO). If the booking is paid, the staff can approve or cancel it. Canceled bookings free up a room. If the booking is unpaid, it is placed back or enqueue in the queue for further processing.

```
=== Staff Menu ===
1. View All Bookings
2. Check Room Availability
3. Process Next Booking
4. Return to Main Menu
Enter your choice: 4

=== Homestay Booking System ===
1. Customer
2. Staff
3. Exit
```

Figure 3.15 : Staff Return to Main Menu

The staff can return to the main menu of Homestay XYZ by entering the number 3. After that, the staff successfully returned to the main menu.

4 Task distribution

Task Description	Assigned To
• Task Distribution • Outline objective of case study • Develop Code • Complete Meeting Log	Choh Jing Yi
• Finalize system requirements, analysis and design • Design the system • Develop Code	Joanne Ching Yin Xuan
• Documentation of requirements and analysis • Complete class diagram and use case diagram of Homestay XYZ Booking System • Develop Code	Muhammad Afiq Danish bin Mohd Hazni
• Documentation of the problem statement and complete synopsis and objective. • Complete and add a description for all the output for customers and staff. • Finalize all documentation and code • Develop Code	Tan Sheau Jun

Meeting Minutes #1

Date/Time		2 January 2025 2.30pm	
Location		Whatsapp	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		2.30pm	-
Joanne Ching Yin Xuan		2.30pm	-
Muhammad Afiq Danish bin Mohd Hazni		2.30pm	-
Tan Sheau Jun		2.30pm	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1.	Group Name Discussion	The group name is yet to be finalized. The members agreed to propose and finalize the name “Happy”.	Choh Jing Yi
2.	Discussion on topic of case study	The topic of case study chosen is Homestay XYZ Booking System	All
3.	Discuss of Objective and problem statement of the mini project	Discuss the objectives and problem statement of Homestay Booking System before starting the project to provide a clear direction for the project	All
4.	Task Distribution for the	● Documentation objective of case	Choh Jing Yi

	mini project	study, • Complete Meeting Log	
		• Design the system • Finalize System requirement, analysis & design	Joanne Ching Yin Xuan
		• Documentation of requirements and analysis • Complete class diagram and use case diagram	Muhammad Afiq Danish bin Mohd Hazni
		• Documentation the problem statement and complete synopsis and objective • Finalize all documentation and code	Tan Sheau Jun
		Development of code	All
5.	Discussion on tool to draw Class Diagram and use case diagram	All group members agree to use Lucidchart software as our tool to develop the class and use case diagram	Muhammad Afiq Danish bin Mohd Hazni

Meeting Minutes #2

Date/Time		6 January 2025 4.30 pm	
Location		Online Meeting(Google Meet)	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		4.30pm	-
Joanne Ching Yin Xuan		4.30pm	-
Muhammad Afiq Danish bin Mohd Hazni		4.30pm	-
Tan Sheau Jun		4.30pm	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1.	Tracking Progress of each member	Each member provided updates on their assigned tasks and any challenges faced.	Choh Jing Yi
2.	Review of Class and Use Case Diagrams	The class and use case diagrams were presented and feedback was given for improvement.	Muhammad Afiq Danish bin Mohd Hazni
3.	Discussion on Code Implementation	Discussed implementing queue-based operations in	All

		the booking system and distributed coding tasks among members.	
4.	Setting Deadlines for Task Completion	We set deadlines on 8 January to complete the coding and testing phases.	All

Meeting Minutes #3

Date/Time		9 January 2025 11 am	
Location		Online Meeting(Google Meet)	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		11.00 am	-
Joanne Ching Yin Xuan		11.00 am	-
Muhammad Afiq Danish bin Mohd Hazni		11.00 am	-
Tan Sheau Jun		11.00 am	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1.	Review of Code Development	Each member showcased their code contributions and discussed any issues faced during implementation.	All
2.	Testing the Booking System	Each member test the booking class system functionalities such as queue or deque and view booking	All

3.	Final Documentation	Reviewed the documentation progress and assigned final tasks for completion.	Tan Sheau Jun
4.	Presentation Preparation	Discussed key points for presenting the project and assigned presentation roles.	Choh Jing Yi